

## Echelon form, Pivot and Free variable:

The following system of linear equation is said to be in echelon form:

$$\begin{aligned} 2x_1 + 6x_2 - x_3 + 4x_4 - 2x_5 &= 7 \\ x_3 + 2x_4 + 2x_5 &= 5 \\ 3x_4 - 9x_5 &= 6 \end{aligned}$$

That is, no equation is degenerate and the leading unknown in each equation other than the first is to the right of the leading unknown in the preceding equation.

The leading unknowns in the system  $x_1, x_3, x_4$  are called pivot variables.

Except pivot variable the other variable  $x_2, x_5$  are called free variable.

- ① If the number of the variable is equal to the number of the pivot then it has unique sol<sup>n</sup>.
- ② If the number of variables is greater than the number of pivot variables then it has more than one solution.
- ③ If in echelon form of a system of linear equation has a degenerate linear equation then it will have no solution.